

Yuchen Rao

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Portfolio: <https://yuchenrao.github.io/>

EDUCATION

Ph.D. Researcher in Computer Science

08/2024-present

Graz University of Technology, Graz, Styria, Austria

Supervisor: Friedrich Fraundorfer

Master of Science in Robotics

09/2016-12/2017

Northwestern University, Evanston, IL, United States

GPA 3.90/4.00

Bachelor of Science in Mechanical and Electrical Engineering

09/2012-07/2016

China Agricultural University (CAU, a Project 985 University), Beijing, China

GPA 3.88/4.00, Rank: 1/43

PUBLICATIONS

• **FlowObject: Flow Steering for Bridging Generative Priors and Reconstruction Fidelity**

Yuchen Rao, Xuqian Ren, Yinyu Nie, Sayan Deb Sarkar, Biao Zhang, Vincent Lepetit, Friedrich Fraundorfer
arXiv preprint arXiv:2606.19019, 2026

• **ICG-MVSNet: Learning intra-view and cross-view relationships for guidance in multi-view stereo**

Yuxi Hu, Jun Zhang, Zhe Zhang, Rafael Weilharter, Yuchen Rao, Kuangyi Chen, Runze Yuan, Friedrich Fraundorfer

Accepted by IEEE International Conference on Multimedia and Expo (ICME), 2025 (**Oral**)

• **Leveraging Automatic CAD Annotations for Supervised Learning in 3D Scene Understanding**

Yuchen Rao, Stefan Ainetter, Sinisa Stekovic, Vincent Lepetit, Friedrich Fraundorfer

Accepted by The IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR) workshop, 2025

• **ReCaLaB: Reality's Canvas, Language's Brush: Crafting 3D Avatars from Monocular Video**

Yuchen Rao, Eduardo Perez Pellitero, Benjamin Busam, Yiren Zhou, and Jifei Song
arXiv preprint arXiv:2312.04784, 2023

• **PatchComplete: Learning Multi-Resolution Patch Priors for 3D Shape Completion on Unseen Categories**

Yuchen Rao, Yinyu Nie, Angela Dai

Accepted by Advances in Neural Information Processing Systems (NeurIPS), 2022 (**Spotlight**)

RESEARCH & WORKING EXPERIENCE

Research Internship, Huawei London Research Center, London, United Kingdom

11/2022-03/2024

Research in Neural Radiance Fields (NeRF) for human body rendering

- Proposed a fully-differentiable pipeline **ReCaLaB** that learns high-fidelity 3D human avatars from just a single RGB video, additionally added an image-conditioned diffusion model to animate appearance and pose of the 3D with unseen human motion.

Research Assistant, Technical University of Munich, Munich, Bavaria, Germany

01/2021-11/2022

Research in 3D Computer Vision, and 3D Machine Learning; Research advisor: Prof. Angela Dai,

- Proposed a 3D shape completion method **PatchComplete** based on learned effective multi-resolution patch priors, and improves over state of the art in novel-category shape completion by 19.3% in chamfer distance on ShapeNet, and 9.0% for ScanNet

Robotics Software Engineer, Berkshire Grey, Massachusetts, United States

05/2019-12/2020

- Fine-tuned Mask RCNN on a custom dataset containing augmented real and simulated RGB images for grocery objects; achieving 92% accuracy for object instance segmentation during grasping
- Contributed to the development of perception modules for ABB robots for object grasping

Robotics Software Engineer, Otsaw Digital Inc, California, United States

07/2018-05/2019

- Improved, tested and successfully delivered a mobile base navigation system on an Ackermann drive robot equipped with Velodyne Lidar for a customer in Singapore
- Created a recovery method to handle a navigational failure based on obstacle detection

Robotics Software Engineer Intern, Honda Research Institute USA, California, United States 02/2018-07/2018

- Simulated and implemented a system for decluttering a table on a Fetch robot with a Kinect RGB-D camera

TEACHING EXPERIENCE

Teaching Assistant, Graz University of Technology, Graz, Styria, Austria

- Computer Vision Summer 2024 ~ 2026

Teaching Assistant, Technical University of Munich, Munich, Bavaria, Germany

- Master Seminar: 3D Machine Learning Summer 2022
- 3D Scanning & Motion Capture Summer / Winter 2021

SKILLS

- Proficient: C/C++, Python, Linux, GitHub, PyTorch, Tensorflow, Anaconda, Jupyter Notebook, ROS, OpenCV, PCL, CUDA, CMake, Gazebo, Rviz, Autoware, MoveIt!
- Experienced: Docker, Mathematica, MATLAB
- Knowledgeable: Generative Model, Computer Vision/Perception, Machine Learning, Deep Learning, Manipulation, Motion Planning

ACADEMIC & ACTIVITY AWARDS

Outstanding graduate of Beijing, China	05/2016
Outstanding graduate of CAU	05/2016
Excellent Student Award granted by the Ministry of Education with Scholarship, China, twice	09/2013-07/2014
Excellent Student Award with Scholarship, CAU	09/2014-07/2015